

Curriculum Vitae

WEI-CHE HSU

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EDUCATION

Arizona State University

Doctor of Philosophy in Industrial Engineering

Aug 2026 – Present

Tempe, Arizona

National Yang Ming Chiao Tung University

Master of Science in Industrial Engineering and Management

Sep 2021 – Aug 2023

Hsinchu, Taiwan

- Total Credits Earned: 37 | GPA: 3.96/4.3
 - **Relevant Coursework:** Advanced Operations Research (simulation optimization), Linear Programming, Integer Programming & Network Analysis, Stochastic Processes, Stochastic Programming
- Thesis: Simulation-based optimization study of procurement lot sizing problem under demand uncertainty
 - Studied a hypothetical raw material procurement problem with demand uncertainty to explore how stochastic factors affect ordering and inventory decisions
 - Developed a simulation-optimization framework that combines heuristic and gradient-based optimization methods, achieving over a 10% improvement in solution quality on large-scale problems compared with using either approach alone

National Tsing Hua University

Bachelor of Science in Industrial Engineering and Engineering Management

Sep 2017 – Jun 2021

Hsinchu, Taiwan

- Total Credits Earned: 164 | GPA: 3.46/4.3
 - **Relevant Coursework:** Operations Research I & II, Probability Theory, Engineering Statistics, Linear Algebra, Discrete Mathematics
- Minor in Power Mechanical Engineering
 - **Relevant Coursework:** Numerical Analysis (approximation algorithms, error analysis), Engineering Mathematics I (ordinary differential equations), Manufacturing Processes, Applications of the Finite Element Method (graduate-level course)
- Data Science Interdisciplinary Program Certification
 - **Relevant Coursework:** Data Structures, Big Data Analytics & Data Mining (graduate-level; data mining techniques and R-based analytical practice), Smart Data Analytics (data analysis methods for engineering and healthcare applications)
- Academia-Industry Collaboration – Sino American Silicon Products Inc. (SAS Wafer)
 - Developed a data-driven frequency control strategy for air-conditioning systems in wafer manufacturing
 - Built regression models and applied genetic algorithm (GA) & particle swarm optimization (PSO) to minimize energy consumption, which saved 70,000 US dollars per month for one of SAS wafer's foundry
- Deep learning study group
 - Studied mathematical models and the application of deep learning models for a semester
 - Applied deep learning models (ANN) to improve glaucoma early prediction accuracy

WORK EXPERIENCE

Micron Technology – Industrial Engineer

Full-time, Department of Operations Intelligence

Mar 2024 – Jun 2026

Taichung, Taiwan

- Improve the transfer efficiency of the Automatic Material Handling System (AMHS)
 - Optimize carrier routing by balancing transfer efficiency and redundant movement under uncertainty
 - Automate the data collection process and enhance system visibility to support monitoring and analysis
- Conducted a project to optimize tool-step assignment strategies aimed at reducing inter-fab transfers under high loading conditions
 - Developed a simulation model to analyze how alternative assignment patterns influence transfer volume, given capacity limits, process-rate variability, and tool-specific step eligibility
 - Applied a heuristic search method to generate improved assignment configurations and identify path-release strategies that reduce inter-fab congestion without affecting the required wafer output

Micron Technology – Industrial Engineer

Intern, Department of Operations Intelligence

Jul 2022 – Aug 2022

Taichung, Taiwan

- Focused on analyzing AMHS cross-fab transfer causes
 - Identified key factors contributing to transfer inefficiencies and proposed actionable solutions
 - Designed interactive Tableau dashboards to automate data visualization

Vanguard International Semiconductor – Data Analytics Engineer

Intern, Department of Production Controlling

Jul 2021 – Aug 2021

Hsinchu, Taiwan

- Constructed production output forecasting models using big data techniques
 - Recognized with 3rd place in the company-wide intern project presentation

TEACHING EXPERIENCE

Teaching Assistant — Course: Data Structure (Class Size: 60 Students) Feb 2023 – June 2023

Department of Industrial Engineering and Management, NYCU

Hsinchu, Taiwan

- Provided student support on algorithm design, data structures, and programming concepts
- Designed homework and exam questions aligned with course learning objectives
- Graded coding assignments for 60 students and developed an automated grading system using predefined input/output cases to ensure consistent and accurate evaluation

TERM PROJECTS

Total Cost Minimization of Stochastic Inventory System

Sep 2021 – Dec 2021

- Examined a stochastic inventory control problem and evaluated policies for minimizing total system cost under demand/lead-time uncertainty
- Implemented simulation optimization methods to assess policy performance in probabilistic environments
- Compared multiple algorithms and analyzed trade-offs among computational effort (number of simulation runs), solution performance, and robustness under uncertainty

Reliability Measures for Partial Highway Systems in Taiwan

Oct 2021 – Dec 2021

- Constructed a network reliability model to analyze the performance of partial highway systems in Taiwan under potential failure scenarios
- Applied a fast enumeration method via Python to evaluate system reliability measures and identify critical network components

Data Mining Analysis of Toronto COVID-19 Case Patterns & Risk Factors

Mar 2021 – June 2021

- Analyzed COVID-19 case data in Toronto to identify key demographic and behavioral risk factors influencing mortality rates
- Employed data-mining techniques, including association rule, random forest, and decision tree models to extract meaningful patterns from public health datasets

PROGRAMMING & ANALYTICAL TOOLS

Programming Languages: C/C++, C#, Python, R, MATLAB, JavaScript

Database: Microsoft SQL Server

Visualization: Tableau, Microsoft Power BI

EXTRACURRICULAR ACTIVITIES

Pop Dance Club

May 2020 – May 2021

- Organized and led a student dance team, managing training schedules and event performance while fostering collaboration and creativity

NTHU IEEM Basketball Team

Sep 2017 – Jun 2021

- Demonstrated teamwork, discipline, and long-term commitment to group performance